Federated Learning Framework for IoT Intrusion Detection

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***Abstract-* As the Internet of Things (IoT) continues to grow, traditional centralized Intrusion Detection Systems (IDS) struggle with issues like data privacy concerns, slow communication times, and the risk of a single failure point. To tackle these problems, Federated Learning (FL) has become a new approach in decentralized machine learning. This paper explores using an FL framework for IoT security, looking at its main types—Horizontal, Vertical, and Federated Transfer Learning—to see which works best for different network setups**

This study specifically examines the use of federated learning (FL) based intrusion detection systems (IDS) in two key areas: Smart Cities and Healthcare (Internet of Medical Things). In Smart Cities, the framework is designed to secure large, diverse networks such as traffic control systems and energy grids, where the main objective is to ensure fast processing and efficient use of bandwidth. In contrast, in the Healthcare sector, the focus is on meeting strict regulatory standards as well as protecting highly sensitive patient data from wearable devices and clinical monitoring systems. A comparison shows that Smart City implementations emphasize scalability and quick response times, whereas Healthcare applications need more sophisticated privacy protection methods.

Keywords: Federated Learning, Internet of Things (IoT), Intrusion Detection System (IDS), Smart City, Healthcare, Data Privacy, Anomaly Detection.

I. INTRODUCTION

The Internet of Things(IOT) is a network comprised of meters, vehicles, buildings, and other items embedded with electronics software and sensors to exchange data through the internet. Such devices function in diverse sectors, such as smart cities, health (IoMT), and industrial control networks (IIoT). IoT devices are composed of heterogeneous nodes and resource-constrained compared to traditional computing nodes, with limited processing power, memory or battery[1]. This architectural complexity makes them attractive targets for botnets, distributed denial-of-service (DDoS) attacks, and unauthorized data exfiltration[2,3].

Federated Learning(FL) is a novel distributed Artificial Intelligence paradigm that allows training machine learning models on multiple decentralized edge devices, without the need to share raw data. In the most traditional FL framework, local devices (clients) independently train a centralized model using their own local datasets and send only resulting model updates or gradients back to central parameter server. This data-to-model rather than model to data base pattern, protects privacy of data by limiting transmission and processing at a given location in order to conform with strict legislation over data protection such as GDPR[4,5].

Intrusion Detection Systems (IDS) are like a smart security guard for your computer network that looks for "bad guys" who have already passed by your front door locks. While your passwords and encryption methods try to keep hackers from even approaching your door, the IDS is on the inside, monitoring what is going on. It is a "second line of defense" that looks for "bad" behavior. IDS systems generally come in one of two varieties: they look on a "Most Wanted" list for viruses (signature-based), or they learn what "normal" looks like and sound an alarm if something acts weird (anomaly-based)[6].

The FL framework for IoT Intrusion Detection is like a global neighborhood watch, where every "smart" house learns to recognize burglars without sharing anyone else their private videos of their own people. In this framework, every IoT device learns its own local intrusion detection model using its own data to learn how to detect intrusions, such as every smart camera or every sensor in a factory.. Instead of sending its private, sensitive data to a giant central cloud, every device sends only tiny "knowledge updates," or tips on what the new hacks look like, to

a central server. The central server aggregates the tips from thousands of devices to generate a master intrusion detection shield, which it sends back to every device. This produces an extremely powerful real-time intrusion detection system that detects entirely new hacks, yet ensures that every user's private information remains safely private and secure on each user's device.

II. RELATED WORKS

IoT security using Federated Learning started with simple privacy-preserving approaches a few years ago. Today, there are self-adaptive FL frameworks that support dynamic network security challenges and heterogeneous hardware.

Understanding that attacks are not static and patterns may change over time, Rehman et al. (2026) [7] introduce an Incremental Federated Learning approach. Training and testing using the CICIoMT2024 dataset they show that using Long Short-Term Memory models under incremental FL allows the IDS to continually learn from new attack data while retaining knowledge about past attacks. They found that when dealing with non-stationary IoT data, "representative learning" yielded the most consistent results.

Fed-DTCN (2026) [8] proposed an unsupervised framework named Federated Dual Temporal Contrastive Network. Since labeled attack traffic can be difficult to obtain under realistic IoT deployments, their framework utilizes contrastive learning to construct a representation of "normal" traffic. Their dual-encoder architecture learns to differentiate between globally-shared traffic patterns and client-specific patterns to improve performance over existing supervised approaches by a significant margin when detecting unseen zero-day attacks.

As these techniques mature for deployment within future-generation networks, Fares et al. (2025) [9] developed a framework using Tab Transformers for intrusion detection within IoT environments. By replacing traditional NNs with transformer architectures tailored for tabular data, as well as utilizing nature-inspired hyperparameter optimization, they achieved state-of-the-art accuracy against benchmark datasets comprised of high dimensional and multi-modal IoT traffic. Researchers have also noted that FL-based IDS trained for next-generation 6G-enabled smart cities (2025) are capable of obtaining up to 98% test accuracy with 35% less communication overhead than traditional FL systems.
Additionally, Mothukuri et al. (updates: 2024/2025)[10] improved the implementation of GRUs to detect DDoS attacks. The research discusses their emphasis on a "first-of-its-kind" federated SVM model that matches the performance of centralized benchmarks while remaining lightweight enough to run on low-power edge devices.

III. TYPES OF FL

A. *Vertical Federated Learning-*

Multiple parties with distinct knowledge of the same set of users collaborate to train machine learning models in a vertical federated learning (VFL) environment without sharing their relevant data or model parameters. It is a form of cooperative ML that protects privacy. A powerful dimension reduction technique called the VFL approach is introduced in this study to choose significant features from subject to excessive by eliminating superfluous and redundant features. A real-world use case for the VFL technique would be: if a capital adequacy card division advertising agency sought to enhance its ML model by examining the most popular products in online shopping domains[11]. Figure 1 more fully describes the VFL system and provides descriptive information

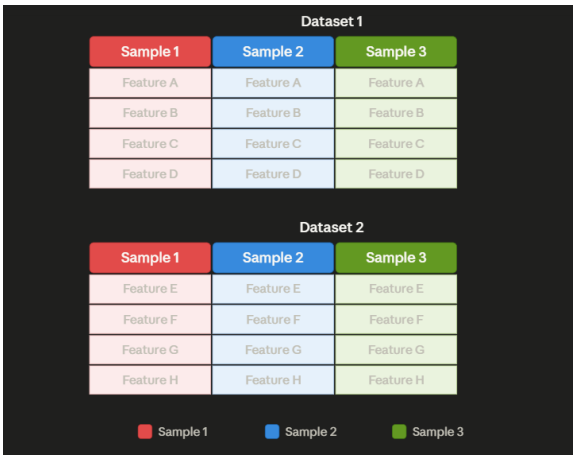

Figure 1

B. *Horizontal Federated Learning-*

When datasets on the gadgets share the very same characteristics with several instances, this is known as horizontal federated learning. The majority of its clients are businesses with consistent sets of data. Consumers who fall under this FL approach share parameters, like area, consumption patterns for some statistical information that was generated, or any other FL result. ML approaches that anticipate the following possible words when a user inputs the text perform more efficiently when given a real stream of data as output from users. The FL method is adroitly applied utilizing aggregated changes from

user activity data without revealing the specific user. Medical images are now considered consumer information and are n't allowed to be participated as a result of restrictions and legislation guarding private health data. FL does, however, allow for the safe transmission of such delicate information through safe aggregated user updates[12] . Figure 2 more fully describes the horizontal FL system and provides descriptive information.

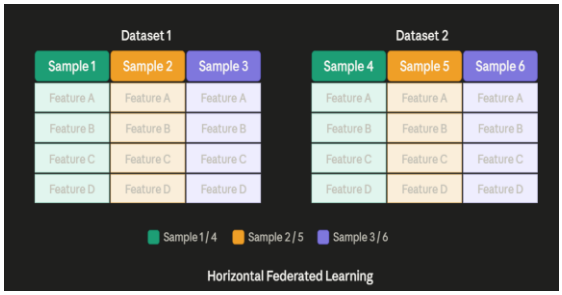

Figure 2

C. Transfer Federated Learning-

Transfer federated learning (TFL) allows different parties to partake scale shows without using the same point space while retaining the while retaining the privacy of their data. In the discipline of machine learning, yields vastly better results than creating a new bone from launch. Real-time examples could be modified rounds of VFL. Participants can use big data and very well ML model statistics to meet their own needs rather than being limited to merely exchanging matching data information[13]. Figure 3 more fully describes the TFL system and provides descriptive information

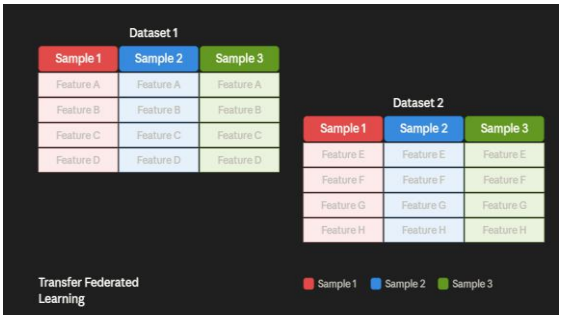

Figure 3

IV. IDS BASED ON FL

Anomaly discovery in original machines is fulfilled by enforcing FL for IDS. DL/ML may now offer unique defensive techniques in a variety of devices as a result of the use of FL for IDS. The security and privacy of such devices are greatly enhanced by intrusion detection systems. The success of ML and DL with IDS in achieving high classification accuracy has given them a lot of traction. Still, the demand to shoot and store data to a centralised server could compromise security and privacy issues. s opposed to transferring data is a decentralized knowledge strategy that protects privacy by training the model locally before transmitting the variables to a centralized server[14]. Moreover, it preserves data privacy, makes the most of bandwidth utilization, and manages an increasing amount of various data more skillfully. It also lessens the computing load on the centralized processing server. FL- grounded NIDS has proven to have significant eventuality for securing the safety of IoT networks. It permits the cooperative learning of an effective intrusion detection system from enormous volumes of traffic data without disclosing private information

V. USE CASE SCENARIO

Any framework's theoretical promise only becomes meaningful when it is put to the test against the intricacy of the real world. Intrusion detection systems based on federated learning are designed to operate in intricate, resource-constrained, privacy-sensitive settings where millions of devices interact simultaneously and where a single security violation might have far-reaching consequences.

To appreciate the practical significance of the proposed FL-IDS framework, it is important to ground it in contexts where its strengths decentralized learning, privacy preservation, and adaptive threat detection are not merely advantageous but genuinely necessary. The vast digital nervous system of a contemporary smart city and the extremely delicate, vital infrastructure of a healthcare IoT network are two such settings that stand out as being especially captivating. Both show environments where a large amount of data must be protected, where there are numerous devices but each one has a finite amount of processing power, and where the effects of an undetected intrusion extend well beyond data loss to encompass human welfare and public safety.

This section shows that FL-based intrusion detection is not a specialised academic exercise but rather a useful, scalable, and urgently needed solution for the IoT security challenges of today by analysing how the suggested framework functions within each of these settings and by making an organised comparison between them.

A. SMART CITY IoT ENVIRONMENT

A smart megacity is, at its core, a living trial in connectivity. A vast and diverse network of IoT bias

connects business signals that react to traffic in real time, waste lockers that alert collection services when they are full, air quality detectors that feed data into public health dashboards, surveillance cameras that monitor public areas, and smart electricity measures that balance grid loads. Smart cities are extremely vulnerable because of this interconnection, which is also what makes them so effective. It is practically difficult for a human administrator or even a conventional centralised intrusion detection system to monitor every data stream in such an environment without causing hazardous bottlenecks due to the sheer volume and diversity of network traffic. Bandwidth is severely strained and unacceptable latency is introduced when all that traffic data is routed to a central server for analysis. More importantly, a large portion of the data produced by smart city devices has implicit privacy consequences that centralised data collecting is unable to sufficiently secure, ranging from citizen mobility patterns to energy usage habits.

In order to overcome these difficulties, the suggested FL-based IDS treats every district or city zone as a separate federated client. Each zone's sensors and edge gateways gather local network traffic and train a lightweight intrusion detection model completely on-device, never sending raw data outside of the local gateway. Every zone periodically communicates with a central aggregate server only its model updates, which are condensed mathematical representations of what the local model has learned. When it comes to the types of threats that smart megacity networks frequently encounter, this armature works particularly well. Attacks known as distributed denial of service(DDoS), which load external systems with unauthorised business, can be linked beforehand on at the zone position before they worsen. Botnet penetration, which happens when compromised IoT bias are secretly signed into malicious networks, can be detected by behavioural anomalies in original business routines. Localised models trained on the business autographs of their own zone can identify man-in-the-middle (MITM) attacks that target communication between bias and gateways. The global model gets better at relating new and changing attack patterns that no single zone might come through on its own because it learns from the concerted experience of all the zones at formerly. Because citizen data noway leaves its zone of origin, this configuration guarantees that smart megacity drivers maintain compliance with data protection fabrics like the General Data Protection Regulation(GDPR) in addition to security performance. The outcome is a security architecture that is both technically sound and morally upright, a feature that is becoming more and more essential to the management of public digital infrastructure.

B. HEALTHCARE IoT ENVIRONMENT

Healthcare reflects the depth and fragility of IoT relinquishment, if smart metropolises represent its breadth. The hospital of today is becoming more and more digital. Patient vital signs are tracked by wearable biosensors. Infusion pumps are ever programmable. Imaging outfit transmits individual data across internal networks. Electronic Health Record(EHR) outstations connect clinicians to patient histories in real time. When combined, these bias produce the Internet of Medical effects(IoMT), a network that's both essential to contemporary clinical treatment and a high target for hackers. Data loss is n't the only metric used to assess the goods of a successful hack in a healthcare IoT setting. Life-saving interventions may be directly delayed by ransomware attacks that help medical labor force from penetrating patient monitoring systems. Years' worth of private patient health information (PHI) may be exposed by unauthorised access to EHR terminals, which would violate both privacy and legal requirements under frameworks such as the Health Insurance Portability and Accountability Act (HIPAA). If the integrity of the very gadgets intended to protect patients is compromised, they may end up being dangerous.

The suggested FL-IDS frame in this script presents each clinic or hospital as a separate affiliated party. Using the business data produced by the sanitarium's internal IoMT bias, intrusion discovery models are trained locally within its own secure network terrain. Case health information noway leaves the hospital's structure — it does n't pass through external waiters, is n't stored in participated depositories, and is n't visible to any reality outside the institution. A federated collaboration server, run by a healthcare institute, a nonsupervisory body, or a dependable third- party aggregator, receives only translated model updates.

This design solves one of the most patient dilemmas in healthcare cybersecurity — the fact that individual hospitals frequently warrant sufficient data to train effective threat detection models on their own, yet are fairly and immorally banned from pooling raw case data with other institutions. This problem is elegantly resolved via federated learning. A small pastoral clinic with limited network business contributes its locally learned model updates to the global aggregation process, and in return benefits from a global model trained on the collaborative experience of dozens or hundreds of institutions. The global model becomes able of detecting pitfalls ransomware propagation patterns, unauthorized device access attempts, anomalous

data transmissions — that no single sanitarium's dataset could have revealed alone.

In this context, the frame's ability to function on resource-constrained bias is inversely significant. Numerous IoMT bias — trendsetter observers, glucose detectors, movable individual tools are n't erected to support heavy calculation. By keeping model training featherlight and ensuring that only compact update vectors are transmitted, the FL- IDS framentegrates naturally into the being tackle geography of clinical surroundings without taking precious structure upgrades.

C. COMPARATIVE ANANLYSIS

Though the Smart City and Healthcare IoT settings both profit deeply from the FL- grounded IDS frame, they represent meaningfully different functional realities. Recognising the diversity of the proposed frame and guiding future deployment opinions in practice both depend on an understanding of these variations.

The two environments are at different ends of the data sensitivity continuum. Because it refers to public infrastructure and aggregate human behaviour rather than to identifiable individuals, smart city data—such as traffic flows, utility usage, and environmental readings—carries mild privacy implications. Healthcare data, by discrepancy, is among the most sensitive information that can live about a person, carrying legal protections, ethical scores, and consequences of exposure that are deeply particular. This difference shapes how aggressively privacy- conserving mechanisms similar as discriminational privacy or secure aggregation.

The trouble biographies of the two surroundings also diverge significantly. Smart megacity networks are most constantly targeted by volumetric attacks similar as DDoS, botnet juggernauts that seek to exploit the sheer number of inadequately secured public bias, and MITM attacks that block dispatches between structure factors. Healthcare networks, on the other hand, are more frequently targeted by financially motivated ransomware, stealthy unauthorized access attempts designed to gather patient data over extended ages, and anomalous transmission actions that gesture data exfiltration in progress. These differences in trouble profile naturally impact how the FL model is trained and streamlined. Given the real- time, high- volume nature of external network commerce, smart municipality deployments profit from frequent model upgrades. Healthcare deployments, where network exertion follows further predictable clinical patterns, can operate effectively with listed, lower-frequence aggregation rounds that put minimum dislocation on clinical operations.
This also showcased using Table 4.

Parameter	Smart City IoT	Healthcare IoT
Device types	Traffic sensors, smart meters, cameras	Wearables, infusion pumps, EHR terminals
Data sensitivity	Moderate	Very high
Primary threats	DDoS, Botnet, MITM attacks	Ransomware, unauthorized access, data exfiltration
Federated clients	City zones / districts	Hospitals / clinics

Figure 4

CONCLUSION

The rapid proliferation of IoT bias has unnaturally converted the threat geography of ultramodern networks, rendering traditional centralized Intrusion Detection Systems decreasingly shy in the face of distributed, privacy-sensitive, and resource-constrained surroundings. This paper presented a comprehensive disquisition of Federated Learning as a principled and practical indispensable — bone that enables IoT bias to collaboratively make robust intrusion discovery capabilities without ever compromising the privacy of the data they induce. The study showed that decentralised model training combined with secure global aggregation can produce discovery performance analogous to centralised approaches while drastically reducing communication outpour and avoiding the problems associated with raw data transmission. It did this by analysing FL types, IDS architectures, and their crossroad. Two precisely examined use case scenario — a Smart City IoT structure and a Healthcare IoT network — were used to validate this frame's real- world connection. The relative analysis between the two scripts further verified that federated intrusion detection is n't a sphere-specific result but a astronomically applicable security paradigm. Future research will probe the implicit incorporation of blockchain for empirical model update auditing, the objectification of discriminative sequestration, and the flexible operation ofnon-IID data distributions. Together, these extensions will further strengthen the foundation laid by this work toward erecting secure, scalable, and privacy-conserving security systems for the IoT ecosystems of hereafter.

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